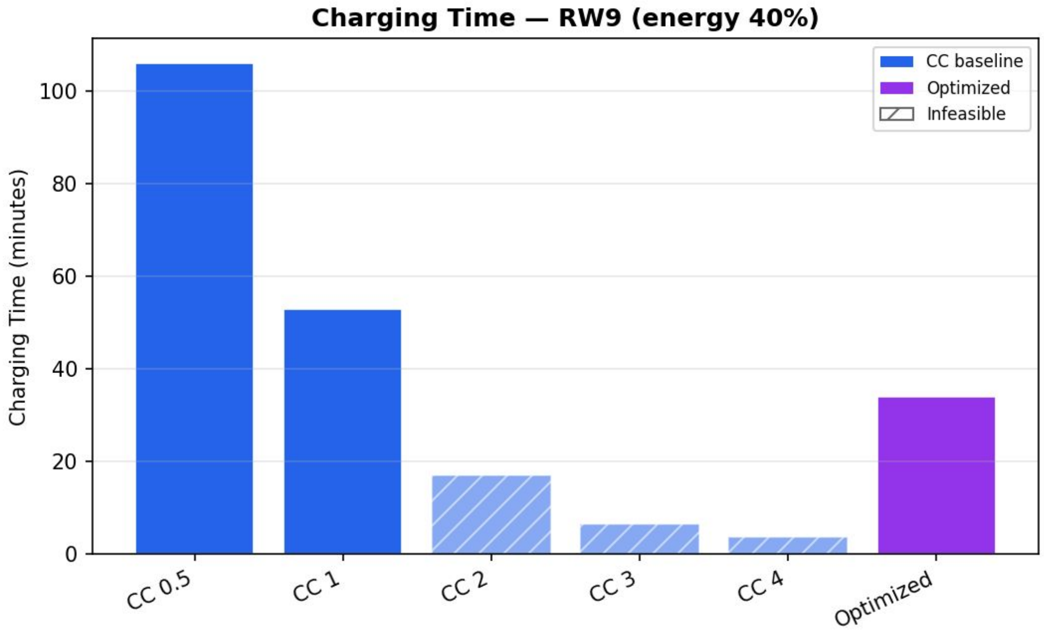
Charging time comparison

Compared with CC 1 A (feasible baseline):

- CC 1 A = 48 min
- Optimized = 33 min
    - 31% reduction in charging time compared with the feasible 1 A constant-current baseline.

Compared with CC 0.5 A:

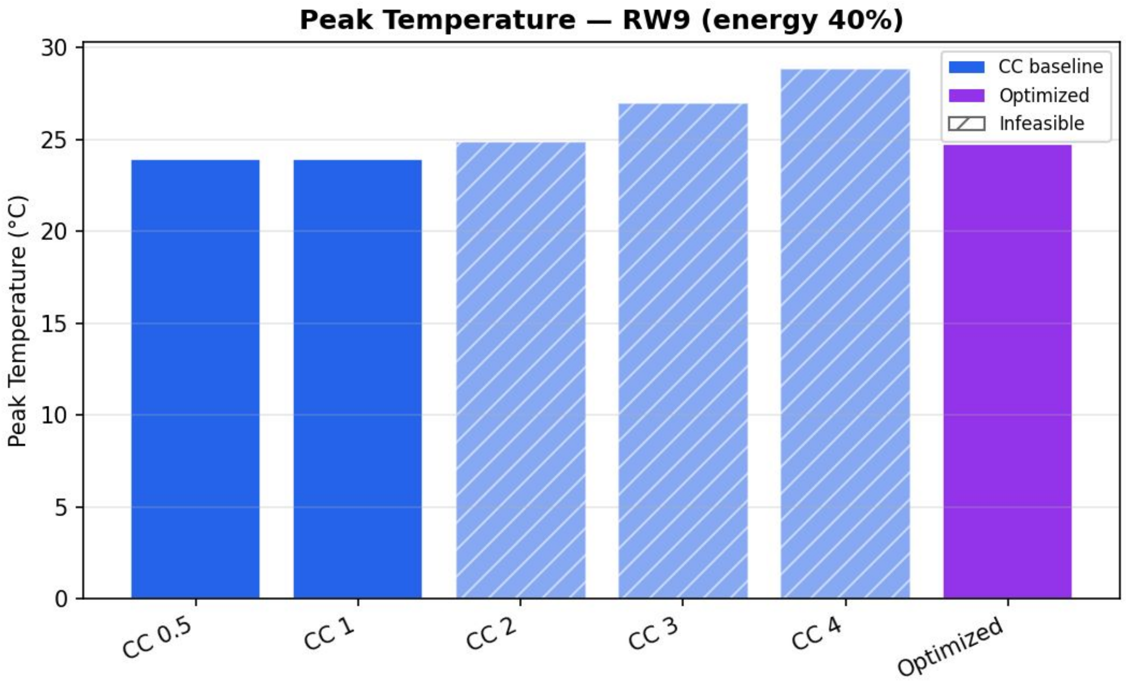
- 66% faster than the 0.5 A baseline.



Temperature comparison

Method	Peak T
Optimized	24.08°C
CC 1	23.92°C

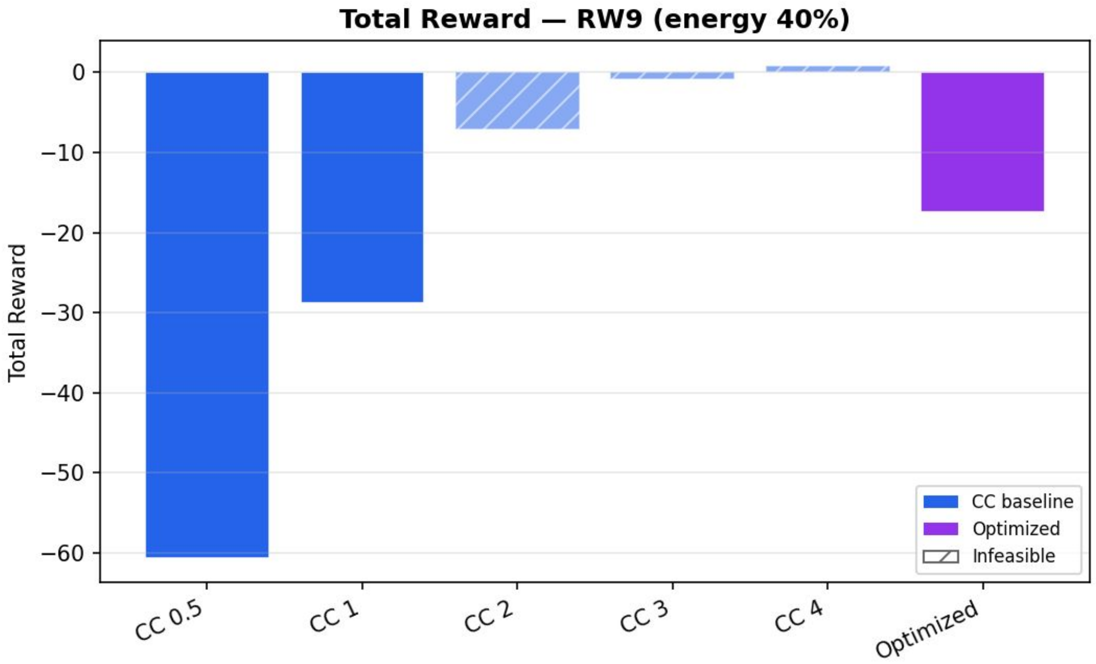
Despite reducing charging time by 31%, the optimized profile increased peak temperature by only 0.16°C compared with CC 1 A.



Reward comparison

Among the feasible charging strategies, the optimized profile achieved the highest objective value.

Higher-current CC baselines appear to have better raw rewards but are infeasible because they terminate before delivering the required energy target.



Observations

Optimization selected a pulsed charging strategy rather than CCCV or multi-step CC.

Compared with feasible CC charging:

- 31% reduction in charging time
- ≈0.16°C increase in peak temperature

High-current constant-current charging (2–4 A) completed much faster, but failed to satisfy the required energy target, making them infeasible under the defined optimization constraint.

Key Results

Metric	**Result**
Selected profile	Pulsed charge/rest
Charging time	**33 min**
Improvement vs CC 1	**31% faster**
Improvement vs CC 0.5	**66% faster**
Peak temperature	**24.1°C**
Δ Peak temperature vs CC 1	**+0.16°C**
Feasible	Yes
Energy target achieved	40%

CONCLUSION

The optimizer automatically selected a pulsed charging strategy.

Compared with the feasible 1 A & 0.5A constant-current baseline, it reduced charging time by about 31% while increasing peak temperature by only 0.16°C.

Higher-current constant-current strategies charged faster but failed to meet the required energy target under our constraints, so they were classified as infeasible.